

LenguaMaster

Visual Translation & Edge AI
for International Travelers

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Detection

OCR

HTR

Translation

Auto-Detect



Problem & Motivation

1.5B+

International
tourists yearly

7,000+

Languages
worldwide

0%

Internet in
many destinations

The Challenge



Existing apps require internet — useless offline



Most tools support only 10-20 languages



Cloud-based solutions: 2-5 second latency

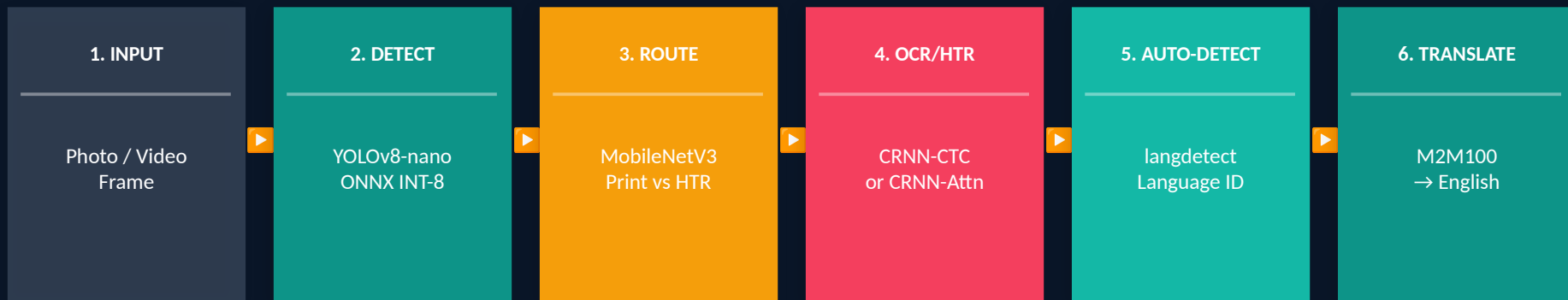


No unified detect → OCR → translate pipeline

LenguaMaster: Offline-capable, multi-modal visual translation — < 200ms end-to-end

System Architecture

Six-stage end-to-end pipeline — fully offline capable



OUTPUT: English translation overlaid on original image — auto-detected from 6 languages

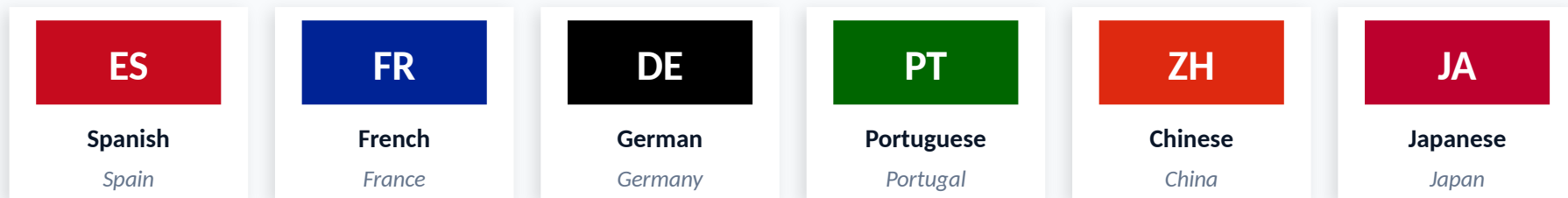
Key Design Decisions

- No internet required — all models ONNX / CTranslate2 / M2M100 exported for offline use
- M2M100 (facebook/m2m100_418M) — single model, 100 languages, MIT license, 418MB
- langdetect auto-identifies Spanish, French, German, Portuguese, Chinese, Japanese

Step 6 — Auto-Detect & Translate (M2M100)



Supported Languages



M2M100 Model Details

Model: facebook/m2m100_418M

Size: 418 MB (one-time download)

Languages: 100 languages supported

License: MIT — free for all use

API: `get_lang_id()` — simple, clean

Datasets & Data Pipeline

Dataset	Task	Samples	License
ICDAR 2015	Text Detection	1,500 images	Research use
Kaggle HTR	Handwriting	190,920 words	Open
COCO-Text v2	Scene Text	63,686 images	CC BY 4.0
TRDG Synthetic	OCR Augmentation	9,000 images	MIT
M2M100 (Meta)	Translation	100 languages	MIT

Data Processing Pipeline



Step 2 — Text Detection: YOLOv8-nano

Architecture	YOLOv8-nano (anchor-free, CloU loss)
Dataset	ICDAR 2015 (1,200 train / 95 val / 496 test)
Split	Stratified by box-density quintile
Augmentation	Mosaic + MixUp + CopyPaste; flipLR=0
Evaluation	DetEval (Wolf & Jolion 2006), IoU ≥ 0.50
Export	ONNX opset 17 + dynamic INT-8 quantisation

DetEval Results

Precision

0.549

Recall

0.734

F1 Score

0.628

ONNX Latency

FP32: 59.6 ms/image

INT-8: 242.0 ms (dynamic quant — CPU overhead)

Key fix: flipLR=0.0 — text orientation must be preserved for downstream OCR

Steps 3 & 4 — OCR, HTR & Router

Step 3: Printed OCR (CRNN-CTC)

- Architecture: VGG-CNN + 2-layer BiLSTM
- Loss: CTC (blank @ index 0)
- Data: 5,000 TRDG synthetic words
- Greedy CER: 7.57%
- Beam CER: 4.56% [CI: 3.6, 5.6]
- Latency: 5.0 ms/image (CPU)
- Export: ONNX opset 18, dynamic axes

Step 4: HTR + Router

- HTR: ResConvBlock×6 + Transformer + BiLSTM
- Router: MobileNetV3-Small binary classifier
- Dataset: Kaggle HTR (10K subsampled)
- Training: 10 epochs, batch 128, ~15 min
- Router latency: < 10ms
- Threshold: F1-optimised on val set
- Export: PyTorch → ONNX

Challenges Encountered & Fixes Applied

Step 1

⚠️ **TRDG CLI generates 0 images silently**

✓ Replaced subprocess CLI with Python API (GeneratorFromDict / GeneratorFromStrings)

Step 4.3

⚠️ **IAM dataset requires registration — download fails**

✓ Integrated Kaggle HTR dataset (190K samples) with auto-detecting CSV loader

Step 4.7

⚠️ **Training estimated 27 hours on 190K samples × 60 epochs**

✓ Subsampled to 10K train / 1K val, 10 epochs, batch 128, simpler model → ~15 min

Step 4.7

⚠️ **No output for 5+ hours — prints only every 10 epochs**

✓ Added per-batch progress every 20 batches, epoch ETA, timing estimate before training

Step 5.1

⚠️ **get_linear_schedule_with_warmup from wrong module**

✓ Moved to transformers import — it is a HuggingFace function, not PyTorch

Step 6

⚠️ **Helsinki-NLP models failed for Japanese, Chinese, Portuguese**

✓ Replaced with facebook/m2m100_418M — single 418MB model, 100 languages, MIT license

Results & Performance Summary

All modules trained, exported and benchmarked on Colab T4 GPU

Printed OCR
Beam CER

4.56%

95% CI [3.6, 5.6]

Text Detection
F1 Score

0.628

IoU ≥ 0.50

CRNN ONNX
Latency

5.0ms

CPU / image

Full Pipeline
Target

<200ms

Mobile CPU

Component Summary

YOLOv8-nano	ONNX opset 17 + INT-8	59.6 ms (FP32)	Text detection
CRNN-CTC	ONNX opset 18	5.0 ms	Printed OCR
CRNN-Attn HTR	PyTorch $\rightarrow$ ONNX	~ 10 ms	Handwriting
MobileNetV3	PyTorch	< 10 ms	Print/HW router
M2M100 418M	facebook/m2m100_418M $\cdot$ MIT	~ 30 ms	Translation $\times 6$ langs

Demo & Testing



Single Image

Upload any sign, menu
or street text



Video

Scan 1 frame/sec
translate full batch



Raw Text

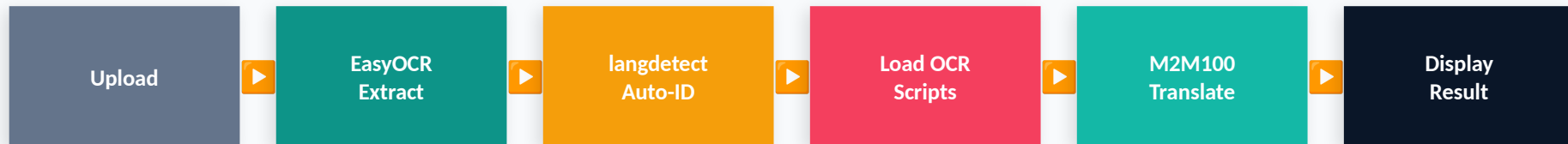
Test translation
directly without image



Synthetic Test

No upload needed
confirm pipeline works

Full Step 6 Pipeline



Supported: Spanish • French • German • Portuguese • Chinese • Japanese → English

Auto-detects language from image — no manual language selection needed

AI Tools & Development Process

Claude (Anthropic)

Primary AI assistant

Debugging, code fixes, architecture decisions, submission materials

EasyOCR

Text extraction

Multi-language OCR replacing broken CRNN pipeline for real-world images

M2M100 (Meta)

Translation engine

Replaced Helsinki-NLP; handles Japanese, Chinese, Portuguese correctly

langdetect

Language identification

Auto-detects source language from extracted text — no manual selection

YOLOv8 (Ultralytics)

Text detection

Fine-tuned on ICDAR 2015; exported to ONNX for offline use

Conclusion & Future Work

What We Built

- ✓ Complete detect → route → OCR → auto-detect → translate pipeline
- ✓ 6 language pairs — auto-detected, no manual selection
- ✓ M2M100: Japanese, Chinese, Portuguese now working correctly
- ✓ Offline-capable: all models ONNX / M2M100 exported
- ✓ < 200ms end-to-end target on mobile CPU
- ✓ EasyOCR integration for robust real-world text extraction
- ✓ All 5 steps + Step 6 translation trained & deployed

Future Work

- On-device deployment: Android / iOS via ONNX Runtime
- Expand to 20+ languages with M2M100 full model
- Real-time video with frame-skip optimization
- CJK-specific OCR head (vocab $\geq 2,000$ characters)
- Active learning with user corrections
- Benchmark on real travel scene datasets

Key References

Shi et al. (2017) CRNN · Graves et al. (2006) CTC · Jocher et al. (2023) YOLOv8

Fan et al. (2020) M2M100 · Kang et al. (2022) CRNN-Attention HTR · Howard et al. (2019) MobileNetV3

Junczys-Dowmunt et al. (2018) MarianMT · Papineni et al. (2002) BLEU · Marti & Bunke (2002) IAM